Foreign Keys in Relational Schemas

txt2tex example

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$$[RecipeId, IngredientId, EmployeeId, CuisineTag, IngredientName, RecipeName, EmployeeName, PositionName]$$

$$
\begin{array}{|l}
FK : \mathbb{P}\ RecipeId \leftrightarrow \mathbb{P}\ IngredientId \\
FKEE : \mathbb{P}\ EmployeeId \leftrightarrow \mathbb{P}\ EmployeeId
\end{array}
$$

Pattern 1 : Single - Attribute Primary Key

Each relation has exactly one pk line. txt2tex underlines the attribute

in the schema box and emits a PK statement immediately after.

Recipe

$$
\begin{array}{|l}
recipeId : RecipeId \\
name : RecipeName \\
cuisine : CuisineTag
\end{array}
$$

PK(Recipe) = $\{recipeId\}$

Ingredient

$$
\begin{array}{|l}
ingredientId : IngredientId \\
name : IngredientName
\end{array}
$$

PK(Ingredient) = $\{ingredientId\}$

Pattern 2 : Composite Primary Key

Two pk lines in one schema body. Both attributes are underlined;

txt2tex collapses them into one PK $set : PK(RecipeIngredient) = \{recipeId, ingredientId\}$.

RecipeIngredient

$$
\begin{array}{|l}
recipeId : RecipeId \\
ingredientId : IngredientId
\end{array}
$$

PK(RecipeIngredient) = $\{recipeId, ingredientId\}$

Pattern 3 : Single - Attribute Foreign Key

A foreign key is a standalone Z predicate. The binding-expression form maps

one source attribute to one target attribute:

(SourceRelation, TargetRelation) $\mapsto \{srcAttr \mapsto tgtAttr\} \in$ FK

RecipeIngredient.recipeId must match a Recipe.recipeId:

$(RecipeIngredient, Recipe) \mapsto \{recipeId \mapsto recipeId\} \in FK$
RecipeIngredient.ingredientId must match an Ingredient.ingredientId:

$(RecipeIngredient, Ingredient) \mapsto \{ingredientId \mapsto ingredientId\} \in FK$

Pattern 4 : Composite FK and Self - Referential FK

Both FK predicates for RecipeIngredient can be stated together.

Each is a separate standalone predicate.

$(RecipeIngredient, Recipe) \mapsto \{recipeId \mapsto recipeId\} \in FK$
$(RecipeIngredient, Ingredient) \mapsto \{ingredientId \mapsto ingredientId\} \in FK$
Employee carries a managerId that references another row in the same relation.

The self-referential FK predicate uses FKEE, declared above.

Employee

$employeeId : EmployeeId$
$name : EmployeeName$
$position : PositionName$
$managerId : EmployeeId$

PK(Employee) $= \{employeeId\}$
Every managerId in Employee must match a valid employeeId in the same relation:

$(Employee, Employee) \mapsto \{managerId \mapsto employeeId\} \in FKEE$